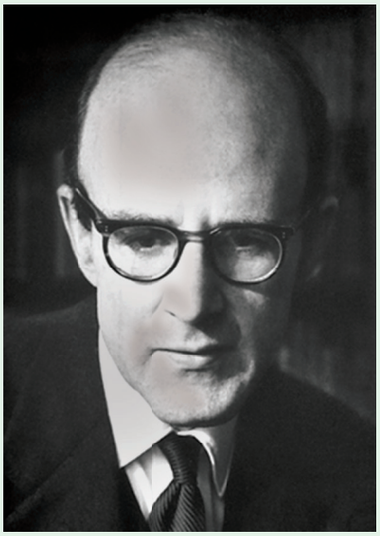
Unit 3 PROTEINS

DJF9NS



Max Perutz

An Austria born British molecular biologist, who first discovered the three dimensional structure of the protein haemoglobin. He was awarded nobel prize for his work in 1962.

Scientific career Fields

Molecular biology
Crystallography

Awards

Nobel Prize for Chemistry (1962)
Wilhelm Exner Medal (1967)
Sir Hans Krebs Medal (1968)
Royal Medal (1971)
Copley Medal (1979)

Learning Objectives

After studying this unit the students will be able to

- Summarize the different physiological and cellular roles of proteins.
- Identify the different dietary sources of proteins
- Describe the relation between amino acids and proteins
- Describe the structure and properties of amino acids
- Identify the pK_a and pI values of amino acids when provided with a titration curve
- Explain the formation and properties of a peptide bond
- Describe the different hierarchical levels of the three dimensional structure of a protein
- Provide structure function relationships for hemoglobin and collagen
- Describe the process protein denaturation
- Relate different diseases with protein deficiency, malfunction and misfolding.

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